A. Discuss Briefly:

1. What is an algorithm, and how is it related to the tasks you perform in your daily life?

Algorithm is a step-by-step process to do something. We use algorithms in our daily life without even realizing that we use them. It helps us to do something that requires selection, sequence, or repetition. An example of this is doing the dishes, where we usually follow a certain order, such as washing the cups first, then the plates, and then the other dishes.

2. Why is it important for an IT student to learn algorithms?

It's important mainly because technology will always have algorithms, just like shown in the video, where there are things that we don't know use an algorithm. And one of the main things an IT student does is create code, where a code has a process that we call an algorithm.

B. Convert each JavaScript code into a step-by-step algorithm using simple English or pseudocode.

Guide: Describe what the program does from the beginning until the loop stops. You may use numbered steps.

1.

```
for (let i = 1; i < 10; i *= 2) {
  console.log(i);
}
```

1. Start with the value of i equal to 1.

2. Check if i is less than 10.

3. If it is, display the value of i.

4. Multiply i by 2.

5. Check again if i is less than 10.

6. Repeat the process until i is no longer less than 10.

7. Stop the loop.

2. **for (let i = 0; i < 10; i++) {**

console.log(i);

}

1. Start with the value of i equal to 0.

2. Check if i is less than 10.

3. If it is, display the value of i.

4. Increase i by 1.

5. Check again if i is less than 10.

6. Repeat the process until i reaches 10.

7. Stop the loop.

C. Run both JavaScript codes. Based on the outputs and execution of the two codes, answer the following:

1. What did you observe about the sequence of numbers produced by the first code?

The first code produces the numbers 1, 2, 4, and 8. I observed that the numbers keep getting bigger by multiplying the previous number by 2 each time.

2. What did you observe about the sequence of numbers produced by the second code?

The second code produces the numbers 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9. I observed that the numbers increase by 1 each time until it reaches 10, which is when the loop stops.

3. What is the main difference between the way the two loops progress?

The main difference is that the first loop multiplies the number by 2 every time it repeats, while the second loop only adds 1 to the number every time it repeats.

4. How does the way an algorithm performs its steps affect its output?

The way an algorithm performs its steps affects its output because different steps or processes can give different results. In the two codes, the first one multiplies the number by 2, while the second one adds 1, which is why they produce different sequences of numbers.